

CoolDesk

AC Service Management Portal

UI / UX Specification Document

Version	1.0 — aligned with User Flows v1.1 & Schema v2.2
Audience	Product, Design, Engineering, QA
Scope	All 4 roles · 19 modules · Full wireframe annotations
Organisation	By role: Owner · Office Staff · Technician · Dealer
Source Documents	User Flows v1.1 PostgreSQL Schema v2.2

0. How to Read This Document

This spec translates the CoolDesk User Flows v1.1 and PostgreSQL Schema v2.2 into actionable UI/UX design requirements. Each screen section follows a consistent structure:

Screen ID	Unique identifier for cross-referencing (e.g. OW-01, OS-03, TC-02, DL-01)
Role	Which actor accesses this screen
Trigger	What action or system event causes this screen to appear
Layout Guidance	High-level layout description — page type, panels, key regions
Field Annotations	Full field table: name, type, required, validation, UI notes
State Behaviour	How the screen changes across status or role conditions
Edge Cases	Specific error states, warnings, and fallback UI that must be designed

Notation Used

 Edge Case	A validation error, failure state, or exceptional UI branch that must be implemented
 Amber Alert	An in-product alert card raised for needs_revisit or chronic jobs — distinct visual treatment required
 Resolved	A previously ambiguous behaviour now confirmed by Schema v2.2 — implement as described
[DB]	Field maps directly to a database column — value must match schema type exactly
[SYSTEM]	Value is computed or written by the backend — UI displays only, does not allow editing
[CONFIG]	Value is sourced from system_config — may change at runtime

Pending Product Decisions

Note: Sections 17.3, 17.5, and 17.7 from the User Flows document contain unresolved product decisions and are excluded from this spec. No UI must be designed for these areas until product owner sign-off is obtained.

1. Global UI Conventions

These conventions apply across every screen in the system regardless of role.

1.1 Authentication & Access Control

All routes are protected by JWT-based role guards. The UI must:

- Redirect unauthenticated users to the login screen immediately
- Render only actions the current role is permitted to perform (see Capability Matrix below)
- Never display disabled versions of unauthorised actions — omit them entirely
- Dealers authenticate via a separate dealer portal with `dealer_credentials` — not the main users table

1.2 Role Capability Matrix

Use this matrix to determine which controls appear on shared screens (e.g. Job Detail).

Capability	Owner	Office Staff	Technician	Dealer
Create jobs	✓	✓	—	✓
Assign technicians	✓	✓	—	—
Advance job status	✓	✓	✓	—
One-state rollback	✓	✓	—	—
Full status override	✓	—	—	—
Cancel job directly	✓	✓	—	—
Submit cancel request	—	—	—	✓
Approve cancel request	✓	✓	—	—
Record payment	✓	✓	✓	—
Edit payment method	✓	✓	—	—
Edit payment amount	✓	—	—	—

Capability	Owner	Office Staff	Technician	Dealer
Void / retain payment	✓	—	—	—
Reopen cancelled job	✓	—	—	—
Manage brands / methods	✓	—	—	—
View analytics	✓	✓	—	—

1.3 Optimistic Locking (HTTP 409 Handling)

Both jobs and payments use version counters. When a conflict occurs:

- Display a non-blocking toast: "This record was updated by someone else. Refreshing..."
- Auto-fetch the latest version and re-render the screen
- Do not discard the user's pending changes silently — show a diff prompt if changes were in flight

1.4 Offline / Sync State (Technician PWA)

The technician interface is a PWA with local queue support:

- Show a persistent "Offline" banner when network is unavailable
- Display a sync queue indicator (count of pending transitions)
- On sync success, refresh job state from server and dismiss indicators
- If server rejects a queued transition (e.g. job cancelled while offline), show error toast with reason

1.5 60-Second Undo Window

After any status transition by a technician, a 60-second undo countdown is displayed:

- Show as a dismissible banner or floating chip at the bottom of the screen
- Display a live countdown timer (e.g. "Undo — 58s remaining")
- Tapping Undo triggers immediate local cancellation if not yet synced; otherwise sends rollback to server
- After 60 seconds, the banner auto-dismisses; the transition is permanent

1.6 Notification Centre

Delivery	In-app only. No push notifications.
Badge	Unread count shown as a badge on the nav bar notification icon
Read state	Clicking a notification marks it as read (is_read = TRUE, read_at = NOW()) and navigates to the relevant job
Deduplication	Backend deduplicates via 4 partial indexes — UI should not implement its own dedup logic

1.7 Status Chip Design

Job status must be displayed as a coloured chip/badge throughout. Suggested palette:

Status	Chip Colour (suggestion)	Notes
pending_schedule	Grey	Dealer-submitted only, awaiting scheduling
scheduled	Blue	—
assigned	Indigo	Active assignment exists
acknowledged	Teal	Technician confirmed
in_transit	Cyan	—
in_process	Amber	Work in progress
completed	Green	Terminal — payment recorded
new	Grey	Complaint only, no scheduled state
resolved	Green	Terminal
needs_revisit	Amber + border	Amber alert must also appear on dashboard card
revisit_scheduled	Blue	Amber alert cleared
resolved_on_revisit	Green	Terminal
cancellation_requested	Red-light	All technician action buttons disabled
cancelled	Red / Dark grey	Terminal — soft-delete NOT applied

1.8 Form Validation Conventions

- All required fields show a red asterisk (*) in the label
- Inline validation fires on blur, not on every keystroke
- Form submit button is disabled until all required fields pass client-side validation
- Server-side errors (e.g. DB constraint violations) surface as field-level error messages below the input
- Cancellation reason, void reason, and rollback reason fields are always mandatory — the submit button must remain disabled until these are filled

2. Owner Role

The Owner has the highest privilege level. All screens available to Office Staff are also available to the Owner, plus the exclusive capabilities listed here. This section covers Owner-exclusive screens and the Owner-specific behaviour on shared screens.

OW-01 · Owner Dashboard

Screen ID	OW-01
Role	Owner
Trigger	Login or navigation to root "/"
Layout	Full-width dashboard with KPI cards row, active jobs table, amber alert panel, and analytics summary

KPI Cards

Top row of metric cards. Each card links to a filtered job list view.

Card	Data Source & Behaviour
Total Active Jobs	Count of jobs WHERE status NOT IN (completed, resolved, resolved_on_revisit, cancelled). Clicking navigates to filtered job list.
Jobs Pending Schedule	Count WHERE status = pending_schedule. Prominent if > 0. Clicking navigates to Pending Schedule queue (OS-06).
Amber Alerts	Count WHERE status = needs_revisit. Displayed with amber background. Zero state hides the card or shows green.
Chronic Jobs	Count WHERE is_chronic = TRUE AND status NOT IN terminal states. Clicking navigates to filtered list.
Today's Revenue	Sum from analytics_business_daily WHERE date = TODAY. Formatted as currency.
No-Show Count	From analytics_business_daily. Links to filtered job list with visit_outcome = no_show.

Amber Alert Panel

■ **Amber Alert:** Show a dedicated "Needs Revisit" panel on the dashboard listing all jobs with status = needs_revisit. Each row shows: Job ID, customer name, brand, assigned technician, date of last

visit, revisit sequence number. Chronic jobs (`is_chronic = TRUE`) show a "CHRONIC" tag. Panel is absent when `count = 0`.

State Behaviour

- Dealer-submitted jobs show a "Via Dealer" tag on the job list
- `is_repeat` and `is_frequent` flags shown as subtle tags on the job card
- Notification badge on nav icon increments for all owner-targeted notification events

OW-02 · Job Detail — Owner View

Screen ID	OW-02
Role	Owner (shares base with OS-05 but with additional controls)
Trigger	Clicking any job from dashboard, job list, or notification
Layout	Two-panel: left = job info + timeline, right = action panel with role-contextual buttons

Header Strip

Field	Type	Req?	Validation	UI Notes
Job ID	UUID (display)	Auto	—	Show shortened (first 8 chars) with copy icon
Job Type	Chip	Auto	—	"Installation" or "Complaint" — colour-coded
Source	Chip	Auto	—	"Direct" or "Via Dealer" — show dealer name if <code>via_dealer</code>
Status	Status Chip	Auto	—	Full status label. Updates in real-time on websocket or poll.
Brand	Text	Auto	—	Brand name + colour swatch from <code>brands.colour_hex</code>
<code>is_repeat</code>	Tag	[SYSTEM]	—	Show "Repeat" tag if TRUE. Tooltip: window used.
<code>is_frequent</code>	Tag	[SYSTEM]	—	Show "Frequent" tag if TRUE.
<code>is_chronic</code>	Tag	[SYSTEM]	—	Show "CHRONIC" tag in red if TRUE.

Customer Panel

Field	Type	Req?	Validation	UI Notes
Customer Name	Text	Yes	—	Editable by owner/staff via separate edit flow
Primary Phone	Tel	Yes	—	Linked from customer_phones. Show all linked numbers.
Address	Text	Yes	—	—
VCID	UUID display	[SYSTEM]	—	Show with link to customer profile view

Owner-Exclusive Action Panel

The following controls appear only for Owners, not Office Staff:

Action	Conditions & UI Behaviour
Full Status Override	Always visible for Owner. Opens a modal with a status dropdown (all valid statuses for job type), mandatory reason textarea, and confirm button. Writes status_rollback timeline event.
Reopen Cancelled Job	Visible only when status = cancelled. Button label: "Reopen Job". Opens modal requiring mandatory reopen reason. On confirm, job returns to last pre-cancellation status.
Edit Payment Amount	Visible when payment exists. Opens inline edit on amount field. Disabled once job is in a terminal state and closed via standard path. Writes payment_edited timeline event.
Void Payment	Visible only when Owner initiates a status reversal outside the 60s undo window and a payment exists. Modal presents two options: Retain or Void. Void requires mandatory void_reason textarea.
Retain Payment	Same trigger as Void. Selecting Retain updates retained_note on the payment. No amount change.

Edge Cases

⚠ Edge Case: When Owner reverses a completed/resolved job status, a mandatory Void-or-Retain modal MUST intercept the action before the status update is applied. This modal cannot be dismissed without a choice.

⚠ Edge Case: Reopen requires querying job_timeline to find the last non-cancelled status. If no prior non-cancelled status exists (edge edge case), show an error: "Cannot determine pre-cancellation status — contact support."

OW-03 · Analytics Dashboard

Screen ID	OW-03
Role	Owner (also available to Office Staff — see OS-09)
Trigger	Navigation to "Analytics" in sidebar
Layout	Tabbed view: Business Technician Scorecards Brand Dealer

Business Tab — analytics_business_daily

Metric	Display & Filtering
Total Jobs	Line chart over date range. Filterable by date range picker.
Completion Rate	Installations completed vs total installations. Cancelled jobs excluded from denominator.
Revenue	Bar chart by day. Total + breakdown by direct vs dealer.
On-Time Rate	Percentage card sourced from on_time_rate column.
Repeat & Frequent Complaints	Count cards. Clicking drills into filtered complaint list.
Avg Resolution Time	In minutes. Displayed as formatted duration (e.g. "1h 42m").

Technician Scorecards Tab

One card per technician showing: jobs_assigned, installations_completed, complaints_resolved, no_show_count, on_time_count, late_count, avg_acknowledgment_mins, first_visit_resolved, avg_star_rating (if review_mode ≠ off).

- Sortable by any metric column
- avg_star_rating column hidden when customer_review_mode = off in system_config

Brand Tab — analytics_brand_daily

- Table view: brand name | installations_completed | complaints_received | complaints_resolved | revenue | revisit_count
- Date range filter applies across all tabs

Dealer Tab — analytics_dealer_daily

- Table view: dealer name | jobs_submitted | complaints | installations | resolved_count | pending_count | avg_resolution_mins

OW-04 · System Configuration

Screen ID	OW-04
Role	Owner only
Trigger	Settings → System Configuration
Layout	Single-page settings form with grouped config items

Configuration Fields

Field	Type	Req?	Validation	UI Notes
repeat_complaint_window_days	Number	Yes	Integer > 0	Label: "Repeat Complaint Window (days)". Default: 30.
frequent_complaint_threshold	Number	Yes	Integer > 0	Label: "Frequent Complaint Threshold (count)". Default: 3.
frequent_complaint_window_days	Number	Yes	Integer > 0	Label: "Frequent Complaint Window (days)". Default: 90.
punctuality_grace_period_mins	Number	Yes	Integer >= 0	Label: "Punctuality Grace Period (minutes)". Default: 15.
customer_review_mode	Select	Yes	off optional mandatory	Label: "Customer Review Mode". Controls Module 19 behaviour.
standard_job_duration_mins	Number	Yes	Integer > 0	Label: "Standard Job Duration (minutes)". Default: 120. Used for

Field	Type	Req?	Validation	UI Notes
				scheduling conflict window.

✓ Changes apply only to new evaluations from the save point onward. Existing flags (is_repeat, is_frequent) are point-in-time snapshots and are not retroactively recalculated.

Edge Cases

⚠ **Edge Case:** Setting customer_review_mode to "off" while jobs are in-progress with pending reviews: the indicator clears automatically on already-completed jobs — no retroactive change needed.

OW-05 · Dealer Account Management

Screen ID	OW-05
Role	Owner only
Trigger	Settings → Dealer Management
Layout	List of dealer accounts + "Add Dealer" button. Clicking a dealer opens a detail/edit panel.

Add Dealer Form Fields

Field	Type	Req?	Validation	UI Notes
business_name	Text	Yes	Non-empty	—
contact_name	Text	Yes	Non-empty	—
email	Email	Yes	Valid email, unique	Used for login identification
region	Text	No	—	Optional geographic region label
Initial Password	Password	Yes	Min 8 chars	Stored in dealer_credentials. Show strength indicator.
Brand Assignment	Multi-select	No	Must be active brands	Picker showing all active brands. Only assigned brands appear in dealer submission dropdown.

Dealer List Actions

- Toggle `is_active`: shown as a toggle switch on each dealer row. Deactivating blocks login immediately.
- Edit brand assignment: add/remove brands from `dealer_brands`. Removing a brand does not affect existing jobs.
- Deactivated dealers show a "Deactivated" badge. Their job history remains accessible.

⚠ Edge Case: Attempting to delete a brand that is linked to a `dealer_brands` row is blocked by DB constraint (ON DELETE RESTRICT). UI should prevent deletion and show: "This brand is assigned to one or more dealers. Remove the brand from all dealers before deactivating."

OW-06 · Payment Method Management

Screen ID	OW-06
Role	Owner only
Trigger	Settings → Payment Methods
Layout	Simple list + Add form

Field	Type	Req?	Validation	UI Notes
name	Text	Yes	Unique, non-empty	E.g. "Cash", "Bank Transfer", "Card"
is_active	Toggle	Yes	—	Deactivated methods are hidden from new job dropdowns but retained in historical payment records (RESTRICT FK)

⚠ Edge Case: Deactivating a payment method that is referenced by existing payments: the historical record retains the `method_id` (RESTRICT FK). The method must not appear in new job payment dropdowns once deactivated.

3. Office Staff Role

Office Staff share most screens with the Owner but lack exclusive controls such as: full status override, reopen cancelled job, edit payment amount, void/retain payment, and system configuration. This section covers Office Staff primary screens.

OS-01 · Office Staff Dashboard

Screen ID	OS-01
Role	Office Staff
Trigger	Login or navigation to "/"
Layout	KPI cards, active jobs list, pending schedule queue banner, amber alert panel, unread notifications

- Pending Schedule queue count displayed as a prominent action card if count > 0
- Amber alert panel mirrors OW-01 — needs_revisit jobs listed with sequence number and chronic tag
- Notification badge shows only notifications addressed to office staff (not owner-only events)

OS-02 · Log New Job — Direct Job

Screen ID	OS-02
Role	Office Staff, Owner
Trigger	"Log New Job" button on dashboard or job list
Layout	Step-form or single-page form: Job Type → Customer Identity → Job Details → Review & Submit

Step 1: Job Type & Source

Field	Type	Req?	Validation	UI Notes
Job Type	Radio / Select	Yes	installation complaint	Determines which fields appear in Step 3. Cannot change after Step 1 without resetting the form.

Field	Type	Req?	Validation	UI Notes
Source	Radio	Yes	direct via_dealer	Selecting via_dealer reveals Dealer selector. chk_dealer_source enforced — dealer_id must be set.
Dealer	Select	Conditional	Must be active dealer	Visible only when source = via_dealer. Dropdown of is_active = TRUE dealers.

Step 2: Customer Identity Resolution

Phone lookup is the primary VCID resolution mechanism. This step drives the linking/creation logic from Flow 1.

Field	Type	Req?	Validation	UI Notes
Primary Phone	Tel input	Yes	Valid phone format	On blur: triggers async lookup against customer_phones. Shows spinner during lookup.
Customer Name	Text	Yes	Non-empty	Pre-filled if phone match found. Always editable.
Address	Text	Yes	Non-empty	Pre-filled if phone match found. Always editable.

Phone Match States

Status	Meaning / UI Behaviour
Match Found (primary)	Show suggestion card: "Link to [Name] — [Address]?" with Confirm / Dismiss buttons. Confirming uses existing VCID. Dismissing creates new VCID and logs customer_link_dismissed event.
Multiple VCIDs matched	Show both matches. Staff must select one. Cannot proceed without selecting or explicitly choosing to create a new record.
Partial match (Levenshtein)	Show secondary suggestion card with lower confidence indicator. Same Confirm/Dismiss logic.
No match	No suggestion shown. Proceeds with new VCID generation.

Step 3: Job Details

Fields shown depend on job type selected in Step 1.

Installation-only fields

Field	Type	Req?	Validation	UI Notes
Brand	Select	Yes	Active brands only	Inactive brand shows error: "This brand is inactive. Please select an active brand."
Scheduled At	DateTime picker	No	Future datetime	If provided, status = scheduled. If omitted, status = pending_schedule (dealer) or triggers error for direct (must schedule at creation for direct).
Unit Details	Repeating row	Yes	At least 1 unit	Each row: Model (text), Unit Type (text), Num Units (number > 0). Add/Remove row buttons.

Complaint-only fields

Field	Type	Req?	Validation	UI Notes
Brand	Select	Yes	Active brands only	—
Issue Description	Textarea	Yes	Non-empty	Minimum 10 characters recommended. No hard DB constraint.
Unit Details	Repeating row	Yes	At least 1 unit	Same as installation.

Step 4: Review & Submission

- Summary panel showing all entered values before final submit
- Submit button labelled "Create Job"
- On success: navigate to the newly created Job Detail screen (OS-05)
- On failure: surface field-level error from server response

Edge Cases

⚠ Edge Case: Brand is inactive: block submission. Error shown on brand field: "This brand is inactive. Select an active brand to continue."

⚠ Edge Case: Dealer source with no dealer_id: Submit button remains disabled. Field-level error on Dealer selector.

⚠ Edge Case: Complaint fields present on installation job (or vice versa): service layer rejects. Show generic error: "Job type mismatch. Please reload and try again."

⚠ Edge Case: Phone number linked to two different VCIDs: both matches surface in the suggestion panel. Staff must manually select one.

OS-03 · Assign Technician

Screen ID	OS-03
Role	Office Staff, Owner
Trigger	"Assign Technician" button on Job Detail (OS-05), or combined with scheduling for pending_schedule jobs
Layout	Modal or right-panel overlay showing technician picker + conflict check result

Field	Type	Req?	Validation	UI Notes
Technician	Select / Search	Yes	Must be active, not deleted	Dropdown of users WHERE role = technician AND is_active = TRUE AND is_deleted = FALSE. Search-filtering supported.

Scheduling Conflict Check

After selecting a technician, the system queries for overlap within the standard_job_duration_mins window. The UI must handle two outcomes:

Status	Meaning / UI Behaviour
No Conflict	Confirm button enabled. Assignment proceeds on confirm.
Conflict Detected	Warning banner shown: "This technician already has job [ID] scheduled in this window." Two options: (1) Acknowledge & Proceed — logs scheduling_conflicts row with canonical ordering. (2) Select Different Technician — returns to picker, no conflict logged.

✓ Conflict INSERT uses canonical pair ordering: job_id = LEAST(A, B), conflicting_job_id = GREATEST(A, B). The chk_canonical_order constraint enforces this at DB level — double-booking is permitted only with explicit acknowledgment.

Reassignment

- If an active assignment already exists, button label changes to "Reassign Technician"
- On reassignment: previous assignment row is marked is_active = FALSE, new row inserted
- New technician receives job_assigned notification
- Previous technician loses the job from their view immediately

⚠ Edge Case: Reassignment while technician is in_transit: permitted by the system. UI should display a confirmation warning: "This technician is currently in transit to this job. Reassigning will remove it from their view. Continue?"

OS-04 · Pending Schedule Queue

Screen ID	OS-04
Role	Office Staff, Owner
Trigger	Navigation to "Pending Schedule" queue, or clicking the KPI card on dashboard
Layout	Filterable table of all jobs WHERE status = pending_schedule

Each row shows: Job ID, Customer Name, Brand, Dealer Name, Submitted At, Days Waiting. Clicking a row opens the scheduling action panel for that job:

Field	Type	Req?	Validation	UI Notes
scheduled_at	DateTime picker	Yes	Must be future datetime	Setting this advances status to scheduled.
Assign Technician	Inline select (optional)	No	Active technician	If selected at the same time as scheduling, two timeline events are written: scheduled → assigned. Conflict check runs if technician is selected.

- Days Waiting counter shown per row — highlight in amber if > threshold (designer to define)
- Empty state: "No jobs pending schedule" with a check icon

OS-05 · Job Detail — Office Staff View

Screen ID	OS-05
Role	Office Staff
Trigger	Any job selection from list, dashboard, or notification
Layout	Mirror of OW-02 with Owner-exclusive controls hidden (status override, reopen, amount edit, void/retain)

One-State Rollback (Staff)

Available for pre-completion states only (in_transit, in_process, acknowledged, assigned). Not available for: completed, resolved, resolved_on_revisit, cancelled.

Allowed from	in_transit → acknowledged, in_process → in_transit, acknowledged → assigned, revisit_scheduled → needs_revisit, assigned → new (complaint) or scheduled (installation)
UI	"Roll Back Status" button in action panel. Opens modal with mandatory reason textarea.
Not allowed	If status is completed, resolved, resolved_on_revisit, or cancelled: button is absent. A message reads: "Status reversal for closed jobs requires Owner access."

⚠ Edge Case: Rolling back from needs_revisit by staff: the revisit record is NOT deleted (unlike the technician 60s undo). The amber alert remains active.

Direct Cancellation (Staff)

Allowed when	Job is in any non-terminal, non-completed state
In-process warning	If status = in_process, show confirmation: "Work has commenced on this job. Confirm cancellation?" before showing reason field.
Field	Mandatory cancellation reason (textarea). Submit button disabled until filled.
After cancel	Job moves to cancelled. Excluded from operational queues. Visible in job history.

Cancellation Request Review (Staff)

When a job has status = cancellation_requested, the action panel shows a "Cancellation Request" badge and the following:

- Dealer's reason is displayed read-only
- "Approve" and "Reject" buttons — both require a decision. No dismiss without action.
- Approving: job moves to cancelled. Dealer notified.
- Rejecting: job returns to its held status. Technician action buttons re-enabled. Dealer notified.

⚠ Edge Case: If the job's lifecycle has advanced while the request was pending, the system auto-rejects it. The UI should poll or use websockets to detect this and refresh the cancellation request badge accordingly.

Payment Method Correction (Staff)

Condition	Job is active (not yet in a closed terminal state). If job is completed/resolved, this control is absent for staff — owner-only.
UI	On the payment panel: "Edit Method" button next to the payment_method field. Opens dropdown of active payment methods.
After edit	Writes payment_edited timeline event with previous and new method values.

OS-06 · Job Timeline Viewer

Screen ID	OS-06
Role	Office Staff, Owner
Trigger	"Timeline" tab within Job Detail screen
Layout	Vertical timeline. Newest event at top. Each event is a card.

Event Card Fields

Field	Type	Req?	Validation	UI Notes
event_type	Label/Icon	[SYSTEM]	—	Human-readable label mapped from enum. Icon-coded by category.
occurred_at	Timestamp	[SYSTEM]	—	Formatted as relative time (e.g. "2 hours ago") with absolute on hover
Actor	Text	[SYSTEM]	—	actor_user_id full_name, or actor_dealer_id business_name, or "System" for system_event
previous_value / new_value	JSON diff view	[SYSTEM]	—	Show as before/after if both present. Format scheduled_at changes as human-readable datetime diff.
reason	Text	[SYSTEM]	—	Shown for rollback, cancellation, void events

- system_event entries styled with a distinct "robot" icon and muted text colour
- "Delayed Sync" system_events show a clock icon with occurred_at from client timestamp

OS-07 · Customer Review — Office View

Screen ID	OS-07
Role	Office Staff, Owner
Trigger	Review tab on completed/resolved Job Detail
Layout	Read-only panel showing review status

- If review_mode = off: review tab is hidden entirely
- If review link was generated: show star_rating (or "Pending" if not yet submitted), review_text, expires_at countdown
- is_low_rated = TRUE: show a low-rating flag on the job card in lists and on the detail header
- is_low_rated = NULL: review not yet submitted — show "Awaiting Customer Response"
- is_low_rated = FALSE: show rating normally without flag

4. Technician Role

Technicians access CoolDesk via a mobile-optimised PWA. The interface is intentionally minimal: a list of assigned jobs and a single job action screen. All transitions are one-tap with a 60-second undo window.

PWA Notes: The technician UI must function offline. Status transitions are queued atomically on-device. The sync queue must be persistent across app restarts (e.g. IndexedDB or service worker cache). Network state must be visible at all times.

TC-01 · My Jobs List

Screen ID	TC-01
Role	Technician
Trigger	App open / home screen
Layout	Scrollable card list of active assigned jobs. Tabs: Today Upcoming All

Job Card Fields

Field	Type	Req?	Validation	UI Notes
Customer Name	Text	[DB]	—	jobs.customer_name
Address	Text	[DB]	—	jobs.address
Job Type	Chip	[DB]	—	"Installation" or "Complaint"
Status	Status chip	[DB]	—	Current status — colour coded per convention
Brand	Text + swatch	[DB]	—	Brand name and colour_hex circle
Scheduled At	Time	[DB]	—	Formatted as "Today 10:00 AM" or "Mon 9 Jun"
Revisit Sequence	Badge	[DB]	revisits only	Shows "Revisit #2" etc. on revisit jobs

- Jobs in cancellation_requested status shown as frozen — grey overlay, no action button
- Empty state: "No assigned jobs" with a checkmark illustration
- Offline banner pinned to top if network unavailable

TC-02 · Job Action Screen

Screen ID	TC-02
Role	Technician
Trigger	Tapping a job card from TC-01
Layout	Full-screen job detail with large primary action button at bottom, undo chip above it

Primary Action Button States

The button label and action change based on the current job status:

Status	Meaning / UI Behaviour
assigned	Button: "Acknowledge" → tapping sets status = acknowledged
acknowledged	Button: "In Transit" → sets status = in_transit
in_transit	Button: "I've Arrived" → sets status = in_process, records actual_arrival = NOW()
in_process (installation)	Button: "Complete Job" → opens payment form sheet
in_process (complaint)	Button: "Mark Resolved" → opens payment form sheet. Secondary button: "Needs Revisit"
revisit_scheduled	Same as in_process flow — Resolved on Revisit or Needs Revisit again
cancellation_requested	No action button. Frozen overlay. Message: "Cancellation pending — awaiting office decision."
completed / resolved / resolved_on_revisit	Read-only view. No action buttons.

60-Second Undo Chip

Appears above the primary action button immediately after any status transition:

- Shows: "Undo — [countdown]s"
- Progress ring or bar visualises the remaining time
- On tap: if entry still in device queue, it is removed locally. If already synced, a rollback is sent to the server.

- On server rollback of Completed/Resolved: payment row is also deleted in the same transaction
- After 60s: chip auto-dismisses. Transition is permanent.

Punctuality Derivation (In Process tap)

When the technician taps "I've Arrived", the backend derives visit_outcome automatically:

- $\text{actual_arrival} \leq \text{scheduled_at} + \text{punctuality_grace_period_mins} \rightarrow \text{on_time}$
- $\text{actual_arrival} > \text{scheduled_at} + \text{grace_period} \rightarrow \text{late}$. late_by_minutes computed and stored.
- UI does not show the outcome to the technician — it is for office/owner reporting only

TC-03 · Payment Form Sheet

Screen ID	TC-03
Role	Technician
Trigger	Tapping "Complete Job" or "Mark Resolved" from TC-02
Layout	Bottom sheet or full-screen overlay presented before final status submission

Field	Type	Req?	Validation	UI Notes
amount	Numeric / Currency input	Yes	amount >= 0	Keyboard type: decimal. Validate before enabling submit.
payment_method_id	Select	Yes	Active payment methods only	Owner-managed list from payment_methods. Deactivated methods not shown.

"Submit" button commits status transition + payment as a single atomic queue entry. Both are sent together and committed in one DB transaction (status → collected).

✓ Payment is inserted directly as "collected" — not as "pending" first. The trg_set_collected_at trigger fires on INSERT and populates collected_at automatically. No second round-trip required.

⚠ **Edge Case:** Technician is offline when submitting: the payment + status transition are queued together as a single atomic entry and synced when connectivity is restored.

⚠ **Edge Case:** Duplicate payment on retry: UNIQUE constraint on payments.job_id prevents double-payment. Server returns 409 which the client treats as success (payment already recorded).

Once submitted, the payment cannot be modified by the technician. The UI must not show any edit controls on the payment panel for the technician role.

TC-04 · Needs Revisit

Screen ID	TC-04
Role	Technician
Trigger	Tapping "Needs Revisit" from TC-02 while status = in_process (complaint only)
Layout	Modal or bottom sheet with reason selector and optional custom text

Field	Type	Req?	Validation	UI Notes
reason	Select (radio)	Yes	Must select one of the enum values	Options: Part Unavailable Customer Not Home Issue Recurring Further Diagnosis Required Other (Custom)
custom_reason	Textarea	Conditional	Required if reason = custom	Visible only when "Other (Custom)" is selected. chk_custom_reason enforced at DB level.

🟡 **Amber Alert:** On submission, an amber alert card is raised on the office staff portal and owner dashboard. This is a system-generated event — no additional UI action is required from the technician.

✓ If is_chronic becomes TRUE on this submission (3rd or more revisit), the system inserts a third_revisit_reached notification to owner + office staff. The technician sees no specific indication of this.

60-Second Undo After Needs Revisit

- Undo chip appears as usual
- If undone: revisit row is DELETED, amber alert is cleared, job reverts to in_process
- After 60s: the revisit row and amber alert are permanent until office staff schedule the revisit

TC-05 · Customer Review Link Generation

Screen ID	TC-05
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Role	Technician
Trigger	Status transition to completed or resolved_on_revisit, AND review_mode ≠ off
Layout	Bottom sheet or overlay after successful completion transition

Optional Mode Behaviour

- Sheet presented after completion confirmation
- Two buttons: "Generate Review Link" and "Skip"
- Skipping: no customer_reviews row created. Job fully closed.
- Generating: review_token (UUID) created, expires_at = link_generated_at + 48h
- After generation: show the shareable URL with a one-tap "Copy Link" button and a "Share via WhatsApp" / "Share via SMS" deeplink

Mandatory Mode Behaviour

- "Skip" button is absent. Technician must generate the link before closing the sheet.
- "Review Pending" indicator shown on the job card in TC-01 until customer submits or 48h expires
- Expiry auto-clears the indicator — it does not block the job from being treated as closed

5. Dealer Role

Dealers access a separate, scoped portal. They can submit jobs, track their own job history, and request cancellations. They cannot see internal details (VCIDs, technician assignments, payment amounts, or analytics).

DL-01 · Dealer Dashboard / Job History

Screen ID	DL-01
Role	Dealer
Trigger	Login to dealer portal
Layout	Read-only job history table with "Submit New Job" CTA. Status shown using simplified labels.

Job History Table Columns

Field	Type	Req?	Validation	UI Notes
Job Reference	Shortened Job ID	[DB]	—	First 8 chars of UUID. Clickable to open DL-03 read-only detail.
Type	Chip	[DB]	—	"Complaint" or "Installation"
Customer Name	Text	[DB]	—	jobs.customer_name — dealer sees their own submitted data
Brand	Text	[DB]	—	Brand name only (no colour swatch required)
Status	Simplified status label	[DB]	—	See dealer status label mapping below
Submitted At	Date	[DB]	—	jobs.created_at
Actions	Contextual	—	—	"Withdraw" (initial state only) or "Request Cancellation" (beyond initial)

Simplified Status Labels for Dealers

Dealers see user-friendly labels, not the internal enum values:

Status	Meaning / UI Behaviour
pending_schedule	Awaiting Scheduling
scheduled	Scheduled
assigned	Technician Assigned
acknowledged / in_transit / in_process	In Progress
completed / resolved / resolved_on_revisit	Completed
needs_revisit / revisit_scheduled	Revisit Arranged
cancellation_requested	Cancellation Pending
cancelled	Cancelled / Withdrawn

- Dealers do NOT see: VCID details, payment information, technician names, or internal analytics
- Notification badge on nav shows only cancellation_request_outcome events addressed to this dealer

DL-02 · Submit New Job

Screen ID	DL-02
Role	Dealer
Trigger	"Submit New Job" button on DL-01
Layout	Single-page form. Simpler than OS-02 — no VCID resolution UI shown to dealer.

Field	Type	Req?	Validation	UI Notes
Job Type	Radio	Yes	installation complaint	Complaint: status = new on submit. Installation: status = pending_schedule.
Brand	Select	Yes	Only brands from dealer_brands	Dropdown scoped to this dealer's linked brands. No other brands visible.

Field	Type	Req?	Validation	UI Notes
Customer Name	Text	Yes	Non-empty	—
Customer Phone	Tel	Yes	Valid phone format	System runs internal phone lookup. Dealer sees only "New Customer" or "Existing Customer" indicator — no VCID details exposed.
Address	Text	Yes	Non-empty	—
Model	Text	Yes	Non-empty	AC unit model number
Unit Type	Text	Yes	Non-empty	—
Unit Details / Issue Description	Textarea	Yes	Non-empty	Label changes based on job type

Post-Submission State

- Success screen: "Job submitted successfully. Reference: [ID]"
- Job appears in DL-01 as "Awaiting Scheduling" (installation) or active complaint
- Owner + all office staff receive dealer_job_submitted notification in-app

Edge Cases

⚠ Edge Case: Dealer account is_active = FALSE: login is blocked at dealer_credentials authentication. The portal shows a "Your account is inactive. Please contact support." screen.

⚠ Edge Case: Brand not in dealer_brands: brand is absent from the dropdown. Dealer cannot submit a job for a brand they are not linked to — there is no override path.

⚠ Edge Case: Network failure during submission: no partial INSERT is committed. Dealer sees a generic error message. Retry button available. The submit button re-enables for retry.

DL-03 · Job Detail — Dealer Read-Only View

Screen ID	DL-03
Role	Dealer
Trigger	Clicking a job row in DL-01
Layout	Read-only detail panel. No edit controls. Simplified status history.

Visible Fields

Field	Type	Req?	Validation	UI Notes
Job Reference	Text	[DB]	—	Shortened UUID
Type	Chip	[DB]	—	—
Status	Simplified label	[DB]	—	See DL-01 label mapping
Brand	Text	[DB]	—	—
Customer Name	Text	[DB]	—	—
Address	Text	[DB]	—	—
Submitted At	Date	[DB]	—	—
Cancellation Outcome	Text	Conditional	—	Shown if a cancellation request was approved or rejected. Label: "Cancellation [Approved / Rejected]".

Explicitly Hidden Fields

- VCID / internal customer identity
- Technician name or contact details
- Payment amount or method
- Internal job_timeline events
- Analytics or performance data

DL-04 · Job Withdrawal & Cancellation Request

Screen ID	DL-04
Role	Dealer
Trigger	"Withdraw" or "Request Cancellation" action from DL-01 or DL-03
Layout	Modal overlaying the current screen

Withdrawal (Initial State Only)

A dealer can withdraw directly only when the job is at its initial status: new (complaint) or pending_schedule (installation).

Field	Type	Req?	Validation	UI Notes
Withdrawal Reason	Textarea	Yes	Non-empty	Minimum character count recommended. Maps to cancellation_reason in DB.

"Withdraw Job" button confirms. On success: job status = cancelled. Job shows as "Withdrawn" in DL-01.

Cancellation Request (Beyond Initial State)

Once a job has progressed beyond its initial status, dealers must request cancellation via office staff review.

Field	Type	Req?	Validation	UI Notes
Cancellation Reason	Textarea	Yes	Non-empty	Submitted to job_cancellation_requests. Visible to office staff in OS-05.

- Submit button disabled until reason is entered
- On submission: job status = cancellation_requested. All technician action buttons frozen.
- Dealer sees "Cancellation Pending" in DL-01
- Dealer receives in-app notification when outcome is decided (approved or rejected)

After Rejection

- Dealer sees "Cancellation Rejected" status indicator on the job in DL-01
- "Request Cancellation" button reappears — a new request is permitted (pending-only index allows it)

Edge Cases

⚠ Edge Case: Dealer attempts a second cancellation request while one is still pending: the submit button must be disabled and replaced with a message: "A cancellation request is already pending for this job."

⚠ Edge Case: Job's lifecycle has advanced while request was pending (auto-rejection): dealer's notification shows "Cancellation Rejected — job has progressed." The job returns to its active status in DL-01.

6. Shared & Cross-Role Screens

SH-01 · Login Screen

Screen ID	SH-01			
Roles	Owner, Office Staff, Technician (via main app) Dealer (via separate dealer portal URL)			
Notes	Two distinct login entry points. Main portal: authenticates against users table using JWT. Dealer portal: authenticates against dealer_credentials table. Both return JWT with role claim.			
Field	Type	Req?	Validation	UI Notes
Email	Email input	Yes	Valid email format	—
Password	Password input	Yes	Non-empty	Show/hide toggle
⚠ Edge Case: Dealer account is <code>_active = FALSE</code> : after credential validation, show "Your account has been deactivated. Contact your administrator." Do not reveal whether the account exists.				

SH-02 · Job List / Search

Screen ID	SH-02
Roles	Owner, Office Staff (full list) Technician (assigned only) Dealer (own submissions only)
Layout	Filterable, sortable table. Filters sidebar or inline filter chips.

Filter Options (Owner / Staff)

Filter	Options
Status	Multi-select from all <code>job_status</code> values. Default: all active.
Job Type	Installation Complaint All
Source	Direct Via Dealer All
Brand	Multi-select from all brands

Filter	Options
Dealer	Multi-select from all dealers (staff/owner only)
Technician	Multi-select from active technicians (staff/owner only)
Date Range	Date picker for created_at range
Flags	Repeat Frequent Chronic No-Show — checkbox filters
Visit Outcome	On-time Late No-show Rescheduled

SH-03 · Notification Centre Panel

Screen ID	SH-03
Roles	All roles
Trigger	Tapping the notification bell icon in the navigation bar
Layout	Slide-in panel or dropdown from nav. List of notifications newest-first.

Notification Item Fields

Field	Type	Req?	Validation	UI Notes
event_type	Icon + label	[SYSTEM]	—	Human-readable label. E.g. "Job Assigned", "Cancellation Request Submitted"
job reference	Shortened ID	[DB]	—	Shown if job_id IS NOT NULL
created_at	Relative time	[SYSTEM]	—	E.g. "5 minutes ago"
is_read	Visual	[DB]	—	Unread items have a blue dot. Read items are dimmed.

- Clicking any notification: marks is_read = TRUE, read_at = NOW(), navigates to relevant job
- "Mark all as read" action available at top of panel
- Empty state: "You're all caught up" with a check icon

SH-04 · Customer Review Submission (External)

Screen ID	SH-04
Role	Customer (external, unauthenticated)
Trigger	Opening a review link shared by technician (contains review_token UUID)
Layout	Standalone public page. No navigation. Branded with CoolDesk/company name.

Field	Type	Req?	Validation	UI Notes
star_rating	Star selector (1–5)	Yes	$1 \leq \text{rating} \leq 5$	Large, touch-friendly stars. Rating required before text review.
review_text	Textarea	No	—	Optional comment field.

- Page validates expires_at before rendering. If token is expired: show "This review link has expired." — no form rendered.
- On submit: UPDATE customer_reviews with submitted_at, star_rating, review_text. is_low_rated generated column auto-sets.
- Success state: "Thank you for your feedback!" — no further action required from customer.

⚠ Edge Case: Token not found (invalid URL): show "This review link is not valid." with no further information.

⚠ Edge Case: Rating ≤ 2 : is_low_rated = TRUE is set automatically by the generated column. Owner dashboard flags the job. No additional UI action required from the customer page.

Appendix A: Screen Inventory

All screens defined in this document:

ID	Screen Name	Role(s)	Source Flow
OW-01	Owner Dashboard	Owner	Flow 1, 4, 6, 9
OW-02	Job Detail — Owner	Owner	Flow 3, 4, 7, 8
OW-03	Analytics Dashboard	Owner, Office Staff	Flow 13
OW-04	System Configuration	Owner	Flow 12
OW-05	Dealer Account Management	Owner	Flow 11
OW-06	Payment Method Management	Owner	Flow 7
OS-01	Office Staff Dashboard	Office Staff	Flow 1, 4
OS-02	Log New Job	Office Staff, Owner	Flow 1
OS-03	Assign Technician	Office Staff, Owner	Flow 5
OS-04	Pending Schedule Queue	Office Staff, Owner	Flow 2, 4
OS-05	Job Detail — Staff	Office Staff	Flow 6, 7, 8
OS-06	Job Timeline Viewer	Office Staff, Owner	All flows
OS-07	Customer Review — Office	Office Staff, Owner	Flow 10
TC-01	My Jobs List	Technician	Flow 3, 4
TC-02	Job Action Screen	Technician	Flow 3, 4
TC-03	Payment Form Sheet	Technician	Flow 7
TC-04	Needs Revisit	Technician	Flow 4
TC-05	Review Link Generation	Technician	Flow 10
DL-01	Dealer Dashboard / Job History	Dealer	Flow 2
DL-02	Submit New Job	Dealer	Flow 2

ID	Screen Name	Role(s)	Source Flow
DL-03	Job Detail — Dealer	Dealer	Flow 2
DL-04	Withdrawal & Cancellation Request	Dealer	Flow 6
SH-01	Login Screen	All roles	—
SH-02	Job List / Search	All roles	All flows
SH-03	Notification Centre	All roles	Flow 14
SH-04	Customer Review Submission	Customer (external)	Flow 10

Appendix B: DB Field → UI Field Mapping (Key Tables)

jobs table (key columns)

DB Column	UI Label	Notes
id	Job ID / Reference	Show shortened (8 chars) in lists; full UUID in detail header with copy
type	Job Type	"Installation" or "Complaint" chip
source	Source	"Direct" or "Via Dealer" chip
status	Status	Status chip per colour convention in Section 1.7
brand_id	Brand	Brand name + colour_hex swatch
dealer_id	Dealer	Dealer business_name — visible to staff/owner, hidden from dealer view
vcid	Customer Identity	Show in owner/staff view. Hidden from dealer and technician.
customer_name	Customer Name	Editable by owner/staff
address	Address	

DB Column	UI Label	Notes
<code>is_repeat</code>	Repeat tag	Boolean flag → show "Repeat" tag if TRUE
<code>is_frequent</code>	Frequent tag	Boolean → "Frequent" tag
<code>is_chronic</code>	Chronic tag	Boolean → "CHRONIC" tag in red
<code>scheduled_at</code>	Scheduled	Human-readable date+time
<code>actual_arrival</code>	Arrived At	[SYSTEM] — shown in timeline and visit outcome panel
<code>visit_outcome</code>	Visit Outcome	on_time late no_show rescheduled — shown in analytics/detail for owner/staff only
<code>late_by_minutes</code>	Late By	Formatted: "23 minutes late" — shown in analytics only
<code>cancellation_reason</code>	Cancellation Reason	Required for all cancellations — displayed in job detail history
<code>version</code>	Hidden	Used internally for optimistic locking — never exposed in UI

Appendix C: Excluded Items

Scope Exclusion: The following items from User Flows v1.1 are intentionally excluded from this specification pending product owner decisions:

Section	Topic	Decision Required
17.3	Rollback from needs_revisit by Office Staff	Should staff rollback delete the revisit row (like the 60s undo) or leave it as a historical record? Design of rollback handler depends on this answer.
17.5	Dealer Visibility on VCID Conflicts	Should the dealer see any indication that their submitted customer phone is already on file? Current spec shows only "New/Existing Customer" indicator with no VCID details. Confirm with product owner.
17.7	Auto-Rejection of Stale Cancellation Requests	Should auto-rejection be triggered by a background job or a service-layer check on every status transition? The system_event timeline format for auto-rejections must be defined before the timeline UI can render these correctly.